

A comparative analysis of Genetic Algorithms and NSGA-II on the Portfolio Optimisation Problem

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Abstract—This paper contains a comparison between a Genetic Algorithm (GA) and a Non-dominated Sorting Genetic Algorithm II (NSGA-II) on the Portfolio Optimisation Problem, based on the Modern Portfolio Theory proposed by Markowitz (1952, 1956).

We use the real-world data of the S&P 500 index (quarterly returns of top 200 stocks from 2019 to 2024, and top 442 stocks from 2004 to 2016), and show that both algorithms can yield returns above the index, in mixed bull and bear market conditions.

N-point Crossover accelerates algorithm convergence, and by using the Sharpe Ratio, the NSGA-II outperformed most models based on Stochastic Dominance we tested, according to the metrics in Bruni et. al. (2017).

Index Terms—Genetic Algorithm, NSGA-II, multi-objective optimization, Sharpe Ratio, Modern Portfolio Theory, Portfolio Optimization

I. INTRODUCTION

The optimisation of investment portfolios is one of the most studied optimisation problems in the economic spectrum. The idea is simple: investors want to obtain the highest possible return from an investment while minimising the risk of losing their investment. However, the complexity and reactivity of the system make achieving good real-world results challenging.

Given the complexity of the issue, various optimisation techniques have been employed [1], which involve the use of different mathematical models to simulate financial markets, depending on the selected optimisation technique.

Investors often seek to ensure guaranteed returns from an investment, with minimal associated risk; by balancing returns and risk, the problem becomes multi-objective.

Often [2], the strategy is to invest in ETFs (Exchange-Traded Fund) that tracks a major market, such as the S&P 500. According to “accepted wisdom” and measured economic performance [2], over the long term, the risk of losing the investment is nearly zero, with a high likelihood of steady returns. Consequently, ETFs that track the country’s stock index are traditionally regarded as stable and efficient compared to mutual funds or investing in individual stocks [3].

The S&P 500 [4] tracks the “500 leading companies [in the USA] and covers approximately 80% of available market capitalization.”. For our dataset, we took the quarterly return data from 2019 to 2024, with the first 3 years being the draining data, and the last 2 the test.

This paper focuses on the use of a classic Genetic Algorithm (GA) [6] and Non-dominated Sorting Genetic Algorithm II (NSGA-II) [7] to explore and compare the effectiveness of these techniques in optimising investment portfolios. The objective is, by drawing a comparison between the two methods, single-objective (return-only) and multi-objective formulations of the problem, and various parameter sets, to identify the most effective strategy for maximising returns with minimised risk.

The article is organised into five sections. Following this introduction, Section 2 details the issue of portfolio optimisation, the economic elements involved, the mathematical model used by the Genetic Algorithm in the optimisation process and the datasets used in our experiments.

Section 3 contains a description the algorithms involved, the GA and NSGA-II, including pseudocode.

In Section 4, we present and analyse the experimental results obtained. We compare the performance of the generated portfolios against the index, assessing the effectiveness of each algorithm in finding optimal solutions. In this section we also asses portfolio performance using several metrics, and compare against optimisation techniques.

Section 5 contains our conclusions and remarks.

II. PROBLEM DESCRIPTION

The portfolio allocation problem can be interpreted as a resource allocation problem: given a finite set of resources, how can we allocate these resources in order to maximise our return (also named profit) with the lowest risk (also named volatility) of investment loss?

Markowitz [8], [9] was among the first to address this problem. He assumed that investors would only accept a higher associated risk if a superior net return is guaranteed. He elaborated the Modern Portfolio Theory to address this problem.

A. Modern Portfolio Theory

In Modern Portfolio Theory, Markowitz proposed the Mean-Variance model as a solution to this problem. Intuitively, Mean-Variance refers to the ratio between the expected return, which is given by the proportionality of the returns of the portfolio components, and the expected risk, which is actually the variance of this return.

$$E(r_p) = \sum_{i=1}^n E(r_i) \times w_i \quad (1)$$

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n \sigma_{ij} \times w_i \times w_j \quad (2)$$

where w_i represents the proportion of asset i , and $\sum_{i=1}^n w_i = 1$. $E(r_i)$ represents the expected return of asset i , $E(r_p)$ represents the expected return of the portfolio, σ_{ij} represents the covariance between asset i and j , and σ_p represents the standard deviation of the portfolio (and σ_p^2 the covariance).

Thus, according to Modern Portfolio Theory, in order to obtain a portfolio with a low degree of risk, we need to maximise (1) and minimise (2).

B. Sharpe Ratio

William Sharpe [10] proposed in 1966 an indicator derived from Modern Portfolio Theory, which helps us measure the performance of a portfolio. Later, this indicator became very popular in describing portfolio performance and was named the Sharpe Ratio.

The Sharpe Ratio is described by (3), as follows:

$$S_p = \frac{E[R_p] - R_b}{\sigma_p} \quad (3)$$

where $E[R_p] = \sum_{i=1}^n w_i \times E[R_i]$ represents the excess expected return of the portfolio. R_b represents the risk-free return which is constant. For our experiments, we used treasury securities. As for January 1st, 2019, the yield of treasury security at maturity was 2.54%. The risk-free used in our experiments is set to 0.0254. σ_p is the standard deviation of the portfolio's excess return which represents the portfolio's volatility.

The Sharpe Ratio can be interpreted as follows: for one unit of risk Y , the portfolio offers X units of return for the associated risk.

In our attempt to improve on Portfolio Optimisation, we use three problem formulations:

- In the first, we optimise the Sharpe Ratio (equation (3)), using a Genetic Algorithm, as the Sharpe Ratio is a widely-used single-objective combining the two separate objectives of Expected Return and Standard Deviation of Portfolio.
- In the second, we optimise, as separate objectives, the Expected Return (equation (1)) and Standard Deviation of Portfolio (equation (2)), using a NSGA II algorithm.
- In the third, we use NSGA II to optimise all three, as separate objectives: the Expected Return (1), Standard Deviation of Portfolio (2), and Sharpe Ratio (3).

Despite this, the overarching objective - and metric - was to return the highest profit at the end of the analysed period of time. This allowed us to draw direct comparison between multi- and single- objective algorithms.

C. Transaction costs

In reality, investments also come with associated costs. In theoretical models, this cost is referred to as *Transaction cost*. The transaction cost [11] can become complex and may be classified as: *internal transaction cost* and *external transaction cost*. The internal transaction cost refers to the effort expended by the investor in creating the investment portfolio, whereas the external cost refers to the cost paid by the investor for information.

The theoretical model on which our experiments and objective functions are based discounts the existence of transaction costs, and this is a blind spot in our approach.

D. Datasets

In our study, we have employed two datasets, an experimental dataset created by us using available data from Yahoo Finance [12] and a control dataset introduced by Bruni et al. [13]. We chose to assess the performance of the algorithms in relation to the S&P 500 index.

The experimental dataset contains quarterly return of the first 200 stocks of S&P500, ordered by market cap, as it was in March 2024, which are not adjusted for dividends or stock splits. The dataset spans approximately 5 years, from 2019 to 2024.

The training period for the algorithms is over the first 3 years, while the testing period is over the last 2. The time frame chosen is relevant to the experiments. Since 2019-2021 is characterised mainly by a bull market (upward market trend), but the testing period is characterised by a bear market (downward market trend), we test our algorithms in different market conditions than those in which they trained (i.e. the datasplit positions the training data in mostly-bull market conditions, while the testing is mostly-bear, or vice-versa).

After finishing algorithm adjustment, we took an additional control dataset; because we iteratively designed the algorithms according to their performance on the 2019-2024 dataset (specifically by adding problem-specific operators and adjusting hyperparameter values), this additional dataset is needed to - partially - control for the information we leaked to the algorithm through our design choices.

The control dataset comprises weekly returns for 442 assets, adjusted for dividends and stock splits, that make up the index, as well as for the benchmark, spanning from November 2004 to April 2016.

In our analysis, we converted the weekly returns into quarterly returns to examine the average return of both the portfolio and the index on a quarterly basis.

Additionally, we divided this dataset according to the 80/20 rule, using 80% of the weekly returns as training data and the remaining 20% for testing.

III. DESIGN AND IMPLEMENTATION

In this section, we detail the components of the Genetic Algorithm used in our experiments, along with the implementation choices for the NSGA-II.

Chang [14] et al. have shown that the number of companies in an index used in portfolio design impacts portfolio performance. The challenge is to find a balance in the number of companies used, as the risk of a portfolio is also influenced by its diversity.

Thus, the proposed algorithms' aim is to find an allocation of the stocks from S&P500 index that outperforms the index's performance. In the context of the Mean-variance model, the goal is to maximise return while minimising risk, denoted by the variation of the portfolio.

Hyperparameter values were chosen starting from literature-suggested values, and using small-scale experiments for adjustments. The largest impact we have encountered is made by varying the number of cutting points for N-Point Crossover, and this is explained below.

A. Genetic Algorithm

The Genetic Algorithm uses an elitist approach, and the use of N-point crossover accelerates evolution by allowing chromosomes to accumulate multiple mutations.

1) *Solution representation*: As we fix the number of stocks to 200 (or 442 for the second dataset), a chromosome is a fixed-size vector of 200 (or 442) weights, each representing the proportion of capital allocated to a specific stock in the portfolio. Since some of the weights may be 0, the algorithm(s) may choose to invest in a subset of the portfolio.

2) *Initialisation and iteration*: In the first step of initialisation, each chromosome is randomly generated, with each locus receiving an uniform random value between 0 and 1. Then, the values are normalised so they sum up to 1 across each chromosome.

The population size is set to 100 individuals, which is subject to evolution over 500 generations.

3) *Fitness function*: The fitness of an individuals is characterised by the Sharpe Ratio (refer to II.B.(3)). As the Sharpe Ratio combines the two objectives - return and risk - the GA performs an single-objective optimisation, attempting to maximise the Sharpe Ratio: to increase the return and decrease the risk.

We have been careful to fine-tune both the fitness function for the GA and the dominance relationship for the NSGA-II so that the population does not lean towards highly volatile companies, but rather towards stable companies. This approach is designed to balance potential returns with risk management, particularly important in fluctuating market conditions.

4) *Selection*: The selection process is designed to maintain a high selection pressure. Thus, the selection is composed of *elitism* and *binary tournament selection*.

As Giorgio et al. [15] showed that elitism enhances the performance of GAs in multi-objective optimisation, and because the Sharpe Ratio is an single-objective adaptor for an underlying multi-objective problem, we chose to use elitism with a ratio of 10% of population size.

We also save, and re-insert into the population, the single best chromosome. GA operators cannot change it, unless they produce a better candidate solution.

Binary tournament selection utilises two breeding pools, each with a size of $k=15$. These two breeding pools are populated randomly with distinct individuals from the current population. Subsequently, the first parent is chosen based on the criterion of having the best fitness from the first pool. Similarly, the second parent is chosen from the second pool based on the same fitness criterion. We used a higher value of k in order to give chance to select fitter individuals for the breeding pools, ensuring a high selection pressure.

Once the two parents are selected, they undergo the crossover process, thereby producing two offspring. These offspring are added to the new population.

5) *Crossover*: Streichert et. al [16] have shown that *discrete N-point crossover* offer better performances in portfolio optimisation problems. In the implementation of this algorithm, we study the performance of the operator using $N = 1$, $N = 15$ and $N = 30$.

To ensure that the sum of weights of the offspring equals 1, the weights will be normalized.

6) *Mutation*: For the mutation process, we utilised two operators applied to the newly obtained population of offspring.

Mutation operator 1 involves the random selection of two distinct positions in a chromosome and swapping the corresponding genes.

For mutation operator 2, two distinct positions in the chromosome, i and j , are randomly generated. The value of the gene at position i is increased, and the value of the gene at position j is decreased by a small value ε . The chromosome is normalised again.

We use $mp_1 = 0.1$, the mutation probability for operator 1, $mp_2 = 0.01$, the mutation probability for operator 2, and $\varepsilon = 0.0001$.

B. NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) was introduced in the early 2000s. Over time, its effectiveness has been experimentally validated in various multi-objective optimisation problems [17], including those related to investment portfolio optimisation [18].

In the experiments, the standard implementation of NSGA-II was utilised (solution representation and genetic operators were based on our Genetic Algorithm implementation), with two modifications to the dominance relationship between individuals, resulting in NSGA-II O2 and NSGA-II O3.

Let p and q be two individuals, with the partial order relation $p \preceq q$ indicating that p dominates q , and N the size of population.

In the first version, $p \preceq q$ if both of the following conditions hold:

- $\text{return}_p > \text{return}_q$
- $\text{risk}_p < \text{risk}_q$

In the second version, $p \preceq q$ if all of the following conditions hold to establish the dominance relationship:

- $\text{return}_p > \text{return}_q$
- $\text{risk}_p < \text{risk}_q$
- $\text{sharpe}_p > \text{sharpe}_q$

Thus, NSGA-II O2 aims to optimise the objective functions of return and volatility, and has 2 objectives, while NSGA-II O3 also includes the Sharpe Ratio as an additional objective function, and has 3 objectives.

Tournament selection is still utilised, chosen from a pool of size $k = 20$, but the best individual for crossover is determined based on the crowding distance operator.

Algorithm 1 NSGA-II

```

1:  $i \leftarrow 0$  ▷ Current iteration
2: Initialise  $P_i$  with a random population
3: Create population  $Q_i$  from sorted by dominance  $P_i$ 
4: while  $i < T$  do ▷ T is stopping iteration
5:    $R_i = P_i \cup Q_i$ 
6:    $F = \text{fast-non-dominated-sort}(R_i)$ 
7:    $P_{i+1} = \emptyset$ 
8:   Populate  $P_{i+1}$  with the best solution from  $F_j$  until size
     of  $N$  is reached.
9:   Sort  $F$  based on dominance
10:  Fill last slots of  $P_{i+1}$  from  $F_{last}$  until  $N$  is reached
11:  Generate  $Q_{i+1}$  from  $P_{i+1}$  using genetic operators
12:   $i \leftarrow i + 1$ 
13: end while

```

IV. EXPERIMENTAL RESULTS

In this section followings, we will split the discussion of experimental results in two parts: in the first part, we will discuss the performance of algorithms on test data; in the second part, we will use various metrics to asses the efficiency of algorithm and compare it with results from literature.

A. Performance evaluation on test data

The performance comparison between GA and NSGA-II is conducted not only against each other but also against a benchmark, comparing it against the S&P 500 index. Additionally, we will evaluate their performances in comparison to findings from existing research literature. The statistical sample size for each data point is 30.

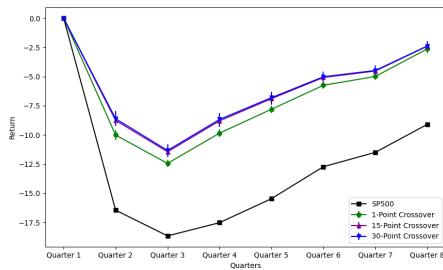


Fig. 1. A comparison of different N for N -point crossover on the experimental dataset

The GA uses a discrete N -point crossover operator. The results highlighted in fig. 1 and 2 suggest that a higher value of N can lead to better convergence towards optimal solutions.

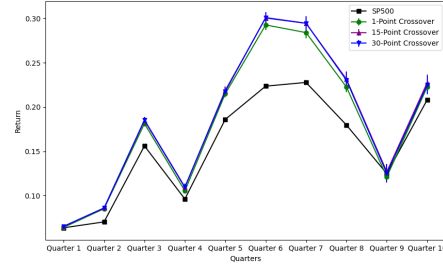


Fig. 2. A comparison of different N for N -point crossover on the control dataset

Additionally, the difference between $N = 1$ and $N = 15$ is relatively significant, whereas the difference between $N = 15$ and $N = 30$ is very small, with later one yielding better results. These findings suggest that increasing the N up to a threshold, the discrete N -point crossover can improve the algorithm's convergences, with minimum improvements for values higher than the threshold.

In the experiments depicted in fig. 3, over the experimental dataset, the sole focus was on maximising the portfolio's returns. Consequently, the GA used the expected portfolio return as its fitness function, and the NSGA-II used the expected return alone as a single-objective function. Both GA and NSGA-II, despite eventually surpassing the index, under performed relative to literature results. This suggests that considering only expected return results in poor allocations. This result enforces the decision of choosing Sharpe Ratio as the fitness function for the GA, and the addition of volatility and Sharpe Ratio as objectives for the NSGA-II.

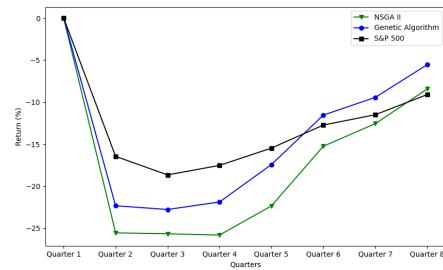


Fig. 3. Performance when maximising only return on the experimental dataset: single-objective optimisation

In fig. 4 and fig. 5 we observe the results obtained by the GA when the Sharpe Ratio is used as the fitness function. It is noticeable that the allocations with the best Sharpe Ratio are among those with the lowest returns for experimental dataset and among highest returns for control dataset.

In fig. 6, the Pareto front found by NSGA-II O2 is presented. It is worth noticing a trend towards solutions that favor a higher return, with the corresponding associated risk. As for control dataset depicted in fig. 7, the results remain

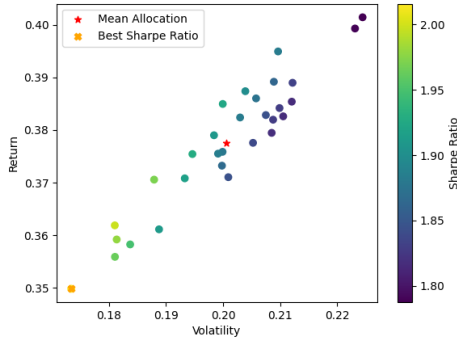


Fig. 4. Results obtained by GA after 30 runs with experimental dataset

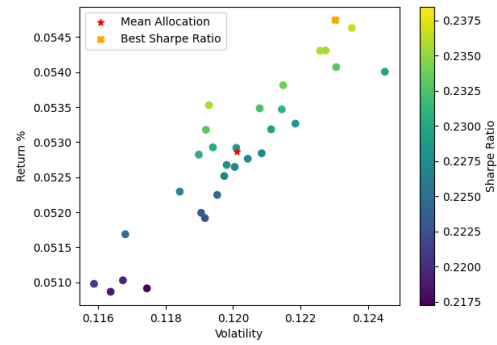


Fig. 7. Pareto front of NSGA-II O2 with control dataset

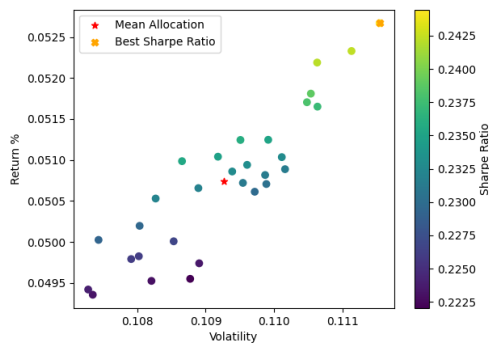


Fig. 5. Results obtained by GA after 30 runs with control dataset

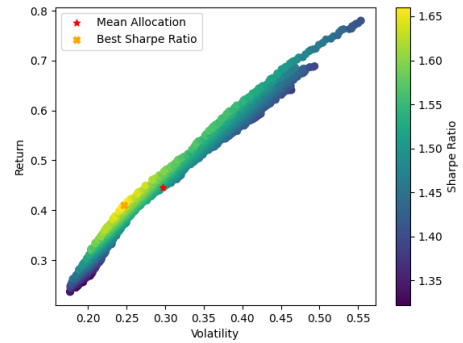


Fig. 8. Pareto front of NSGA-II O3 with experimental dataset

consistent with the previous observation. Moreover, on the control dataset, NSGA-II O2 found the best front composed only of one chromosome, whereas on the experimental dataset, it found many more.

The results obtained by NSGA-II O3 are depicted in fig. 8 and fig. 9. The results suggest that NSGA-II O3 focuses on allocations with much lower volatility than NSGA-II O2.

The inclusion of Sharpe Ratio as objective function helps NSGA-II O3 to prioritise not just higher returns but also the discounted ratio between return and volatility, hence favoring

more balanced investment strategies with lower volatility.

We believe that adding the Sharpe Ratio explicitly, alongside return and volatility, gives the algorithm access to more information about the problem: as the ratio models a relationship between return and risk, the algorithm can take it as a cue, and not rely on the generic non-dominance relationships alone.

In the case of NSGA-II O2, it is clear that the algorithm aims to find allocations with high returns somewhat propor-

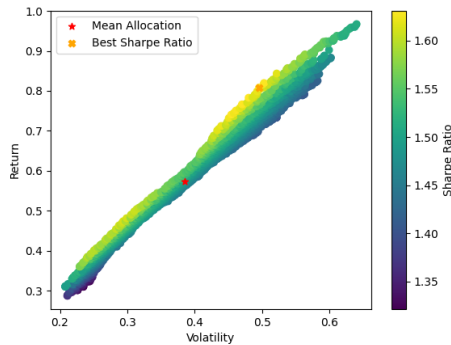


Fig. 6. Pareto front of NSGA-II O2 with experimental dataset

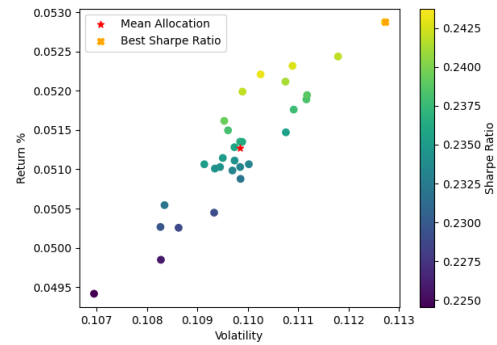


Fig. 9. Pareto front of NSGA-II O3 with control dataset

tional to the risk, because the objectives functions involved are only return and volatility. On the other hand, NSGA-II O3 also takes into account the discounted proportionality between volatility and return (Sharpe Ratio), which indirectly seeks allocations that are more resilient during high volatile periods. This approach in NSGA-II O3 allows for a more balanced strategy, emphasising not just the magnitude of returns, but also the efficiency and sustainability of those returns relative to the risks involved.

Increasing the Sharpe Ratio can be achieved either by increasing the numerator (excess return) or by decreasing the denominator (volatility). In practice, reducing volatility is easier and more controllable than increasing returns, especially when high returns come with proportionally higher risks.

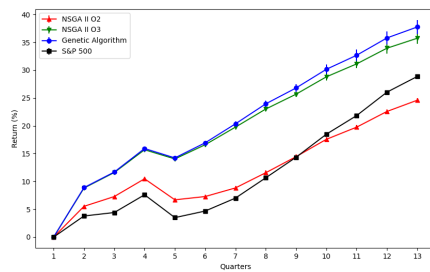


Fig. 10. Performance of S&P500, GA, NSGA-II O2 and NSGA-II O3 on the in-sample experimental dataset

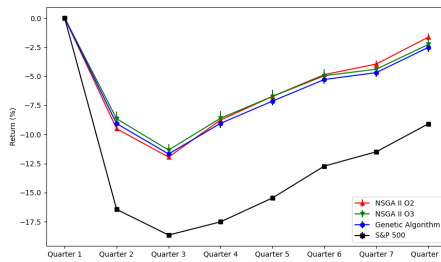


Fig. 11. Performance of S&P500, GA, NSGA-II O2 and NSGA-II O3 on the out-of-sample experimental dataset

These observations are further backed by the results illustrated in fig. 11 and fig. 13, representing out-of-sample results and fig. 10 and 12 representing in-sample results. The data indicates that GA and NSGA-II O3 delivered the best performance, surpassing the index on both the experimental and control datasets over out-of-sample results, with NSGA-II O3 slightly outperforming GA.

In the experimental dataset, NSGA-II O2 displayed the least favorable performance in both out-of-sample and in-sample results, particularly over in-sample results where NSGA-II O2 has fallen below the benchmark. Conversely, in the out-of-sample control dataset, its performance was limited to the first two quarters, after which it lagged behind, even falling below

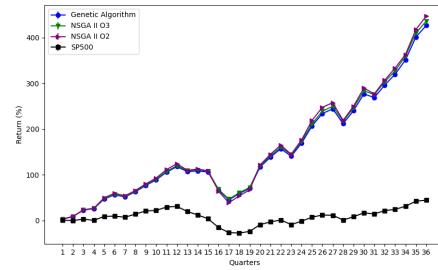


Fig. 12. Performance of S&P500, GA, NSGA-II O2 and NSGA-II O3 on the in-sample control dataset

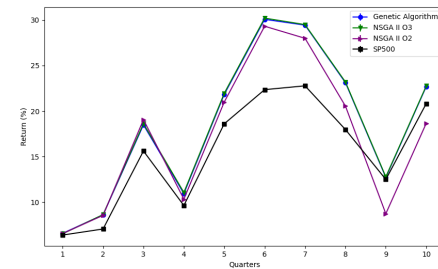


Fig. 13. Performance of S&P500, GA, NSGA-II O2 and NSGA-II O3 on the out-of-sample control dataset

the index in quarters 9 and 10. In case of the in-sample control dataset, NSGA-II O2 performed the best where the trend of market was upward, and falling below NSGA-II O3 and GA wherever a sharp downward trend of the market appears.

The GA model has shown strong performance across both the experimental and control datasets (which contain both bull and bear market conditions) over both in-sample and out-of-sample results, consistently beating the benchmark and NSGA-II O2, except of in-sample control dataset. This performance shows the efficacy of using the Sharpe Ratio as a fitness function in the GA. This strategy proves effective in maintaining stability and achieving better portfolio allocations under varying market conditions (both bull and bear) – in other words, the Sharpe Ratio is responsible for the majority of the performance of our algorithms. NSGA-II O2 can sometimes (fig. 10, 11) equal and slightly surpass it, but this comes at the cost of increased instability, and worse average performance.

NSGA-II O3 consistently delivered the best outcomes across both datasets, over in-sample and out-of-sample results, slightly outperforming the GA and showing significant improvement over NSGA-II O2, especially over out-of-sample results. NSGA-II O3's ability to identify more robust asset allocations highlights its advantage in handling market volatility effectively.

Thus, incorporating the Sharpe Ratio as an objective in NSGA-II and using it as a fitness function in GA, not only favors less volatile assets, but also showcase consistent per-

formance, which is extremely important in financial markets.

B. Comparative efficiency analysis

To gain deeper insights into the performance of our genetic algorithms, we conducted a comparative analysis with the study of Bruno et al. [19].

In their research, Bruni et al. explore portfolio optimisation through a framework of Stochastic Dominance (SD) to achieve Enhanced Indexation. Their methodology centers on constructing portfolios that stochastically dominate the benchmark, employing a variety of both exact and approximate SD models. These models are integrated into a reformulated Linear Programming framework that have a more compact and computationally efficient format than the basic one with exponentially many constraints.

In their analysis, they have adopted a Rolling Time Window (RTW) evaluation scheme with a 52-week in-sample period and a 12-week out-of-sample period for re-balancing. For this specific side-by-side comparison, we have employed their evaluation scheme.

A further evaluation of out-of-sample performance is conducted using several metrics. In the description of metrics, the R and $E[R]$ refer to return out-of-sample in Bruni's analysis, while in ours refers to weekly return. The r_f represents the risk-free factor, which is set to 0 and R^I represents the weekly return of index. The best metric values are written in bold.

The metrics are described bellow as following:

- **Sharpe Ratio**: Initially presented in (3), reproduced here:

$$S_p = \frac{E[R_p] - R_b}{\sigma_p}$$

- **Sortino Ratio** [21]: It is preferable when assessing investments that may have skewed return distributions. The bigger the ratio, the better.

$$S_r = \frac{E[R - R_f]}{\sigma(\min\{R - R_f, 0\})}$$

- **Rachev Ratio** [22]: A ratio that assesses the tail risk by comparing the Conditional Value at Risk (CVaR) at different confidence levels α and β .

$$R_r = \frac{CVaR_\beta(R_f - R)}{CVaR_\alpha(R - R_f)}$$

where α and β are set to 5%.

- **Information Ratio** [23]: Measures the excess return to the volatility of the excess return, giving insights into the consistency of portfolio returns.

$$I_r = \frac{E[R - R^I]}{\sigma(R - R^I)}$$

- **Turnover**: Represents the measure of the portfolio's trading activity, used to estimate transaction costs. Since we do not include transaction costs in our study, this metric is not included in this analysis.

$$\text{Turnover} = \frac{1}{N_{\text{reb}}} \sum_{j=1}^{N_{\text{reb}}} \sum_{i=1}^n |x_{j,i} - x_{j-1,i}|$$

- **Jensen's Alpha** [24]: A measure of the average return of the portfolio above the expected return based on the market risk, adjusted for the cost of capital.

$$\alpha = (E[R] - R_f) - \beta(E[R^I] - R_f)$$

where β is the portfolio's beta coefficient.

- **Average Return**: The simple mean of the weekly return over observation time frame.

$$\text{Average Return} = E[R]$$

Tables 1 and 2 showcases the analysis of metrics achieved by the models used in Bruni's study, including outcomes from NSGA-II O2, NSGA-II O3, and GA. The results show that both the GA and NSGA-II O3 models outperformed nearly all the stochastic dominance approaches applied in the study. However, the LR-ASSD model achieved the best results for the Rachev Ratio, with the KP-SSD model matching its performance for the Jensen's Alpha.

It should be noted that the GA model performed comparably to the NSGA-II O3 model across all metrics, with the exception of the **Average Return**, where there was a notable difference.

This analysis clearly indicates a distinction in model performances across different metrics, underlining the importance of choosing the right model based on the financial outcomes one aims to optimize. While genetic algorithms, in particular the presented GA and NSGA-II, show broad-based strength, the specialized performance of certain SSD models cannot be overlooked, especially in metrics where they exhibit comparable efficacy.

TABLE I
PERFORMANCE METRICS - SHARPE R., SORTINO R., RACHEV R., AND INFORMATION R.

Approach	Sharpe R.	Sortino R.	Rachev R.	Info R.
CZE-SD	0.08	0.11	0.97	0.10
RMZ-SSD	0.09	0.12	0.93	0.04
LR-ASSD	0.09	0.12	1.12	0.07
L-SSD	0.08	0.10	0.98	0.03
KP-SSD	0.08	0.11	1.01	0.06
MeanVar	0.09	0.11	0.98	0.03
Index	0.05	0.06	1.03	-
GA	0.12	0.16	1.11	0.30
NSGA-II O2	0.11	0.15	1.11	0.25
NSGA-II O3	0.12	0.16	1.10	0.30

V. CONCLUSION

In this paper, we have demonstrated the efficiency of Genetic Algorithms in addressing the problem of portfolio optimisation. We have also highlighted the important role of adapting the genetic operators.

In our experiments, the use of the discrete N-point crossover operator shows that choosing the appropriate value of N is crucial for the convergence of the algorithm; a larger N appears to favor faster convergence (as expected, since it allows for faster accumulation of mutations, and quick Neutral Network

TABLE II
PERFORMANCE METRICS - TURNOVER, JENSEN'S ALPHA, AND AVERAGE RETURN

Approach	Turnover	Jensen's	Aver. Return
CZ ϵ SD	1.70	0.0012	0.0026
RMZ-SSD	1.30	0.0012	0.0021
LR-ASSD	1.67	0.0009	0.0019
L-SSD	1.54	0.0003	0.0013
KP-SSD	1.42	0.0022	0.0036
MeanVar	1.12	0.0009	0.0018
Index	-	-	0.0013
GA	-	0.0022	0.0036
NSGA-II O2	-	0.0021	0.0036
NSGA-II O3	-	0.0022	0.0037

[20] escape). However, beyond a certain point, the algorithm tends to get stuck in a local optimum – the accelerated convergence becomes premature convergence.

In addition, in the scenarios analysed, both the standard GA and the NSGA-II variations succeeded in surpassing the benchmark's performance (the index) on the experimental and control datasets. We demonstrated that using the Sharpe Ratio as an objective for NSGA-II is an effective metric for identifying optimal portfolios; the performance of the Evolutionary Algorithms used must, then, reside in their ability to efficiently search through the large space of possible allocations.

In our comparative study with Bruni (2017) et al., both Genetic Algorithms (GA) and their NSGA-II O3 variant demonstrated a notable advantage over almost all studied Stochastic Dominance (SD) based models across proposed metrics. This superiority suggests that the evolutionary nature of GA and NSGA-II, with their capability to dynamically adapt and evolve solutions, can be effective for complex optimization tasks rather than the traditional SD models, which tend to focus more on theoretical dominance relationships in financial datasets.

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